

ONE-SIDED PARITY COMPRESSION FOR CERTIFIED SAT ATTACKS ON THE 33-POINT ERDŐS-SZEKERES PROBLEM

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ABSTRACT. The first unresolved exact case of the Erdős–Szekeres convex-polygon problem asks whether every 33 points in general position contain seven in convex position. Recent work already introduced one convexity variable per four-set and an ordered four-clause signotope encoding. We isolate a narrower compression: a nonconvexity flag is reified only in the direction needed for an UNSAT proof. Under the signotope one-change rule, the required odd-parity condition factors into one endpoint disequality and one middle equality, so the half-reification needs four ternary clauses, compared with sixteen five-literal clauses for the audited public direct full reification and eight ternary clauses for a locally clause-minimal factored full equivalence. For 33 points the resulting refutation-sound relaxation has exactly 46,376 variables, 4,599,408 clauses, and 150,503,760 literal occurrences. Every geometric 33-point counterexample induces a satisfying assignment; hence a checked refutation would prove the conjectured value. We supply a streaming generator, independent truth-table and geometric verifiers, and a checked DRAT-to-LRAT proof pipeline on the known $ES(5) = 9$ instance. We do not produce a refutation of the full 33-point formula or a realizable 33-point counterexample, so this paper does not determine $ES(7)$. We also remove an ambiguity about the 17-point strength test: the theorem of Szekeres and Peters already implies that $F_{17,6}$ is UNSAT over all signotopes. For the geometrically normalized direct-relation encoding, a complete 61-cube partition produces 61 checked DRAT refutations, all converted to and verified again as LRAT.

1. INTRODUCTION

For an integer $k \geq 3$, let $ES(k)$ denote the least N such that every set of N points in the plane in general position contains k points in convex position. Erdős and Szekeres first published the general problem in their 1935 paper [4], crediting its original question to Esther Klein. Their later construction gives

$$ES(k) \geq 2^{k-2} + 1,$$

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and they conjectured equality [5]. The case $k = 6$ was settled computationally by Szekeres and Peters [13]; a subsequent Isabelle/HOL development supplied a new, substantially faster formal proof of the case [9]. Hence $k = 7$ is the first unknown exact case.

At the date of this manuscript, the rigorously justified numerical interval is

$$(1.1) \quad 33 \leq \text{ES}(7) \leq 113.$$

The lower endpoint follows from the classical construction. The upper endpoint is the specialization at $k = 7$ of the bound of Mojarrad and Vlachos [10]:

$$\text{ES}(k) \leq \binom{2k-5}{k-2} - \binom{2k-8}{k-3} + 2, \quad \binom{9}{5} - \binom{6}{4} + 2 = 113.$$

Recent structural results establish the conjectured threshold for split polygons and for decomposable point sets, but not for arbitrary point sets [1]. A 2025 computational study encoded the 33-point case and reported refutations for several convex-layer-anchored subfamilies, together with pronounced heavy-tailed solver behavior on the remaining search [3]. These advances leave the unanchored case open.

SAT methods have already proved useful for order-type problems [2, 11]. Their mathematical value depends on a sharp separation between three statements:

- (i) every geometric counterexample maps to a satisfying assignment;
- (ii) an UNSAT solver run terminates on the entire formula; and
- (iii) the emitted proof trace is accepted by an independent checker.

This paper proves (i) for a smaller encoding, checks a complete proof pipeline on a known exact case, and supplies exact checks of all formula counts. It does not claim (ii) or (iii) for the 33-point instance.

The prior-art boundary is important. Subercaseaux, Mackey, Qian, and Heule [12] already used one convexity variable per four-set, expressed convexity by orientation parity, and used the standard four-clause ordered signotope axioms. Koshelev and Koshka [8] subsequently split convex-quadrilateral constraints into endpoint-equivalence and middle-disequality cases; the complement of that split is the factorization used below. Accordingly, we do not claim novelty for the four-set variable, parity characterization, signotope clauses, or endpoint/middle factorization. Our narrower contribution is their polarity-aware combination for a refutation: one nonconvexity witness per four-set, no reverse implication, four ternary guard clauses rather than the public implementation’s sixteen five-literal full-reification clauses, and uniform integration into all k -set exclusion clauses with exact 33-point accounting. Section 3 proves the local geometric criterion, Section 4 derives and minimizes the ordered signotope CNF, Section 5 proves the refutation-sound direction, and Section 6 gives audited static comparisons. The distinction

between abstract signotope assignments and realizable point configurations is maintained throughout.

2. ORIENTATIONS AND CONVEXITY

Let $P = (p_1, \dots, p_n)$ be a labeled planar point set in general position. For distinct labels i, j, k , put

$$\chi(i, j, k) = \text{sgn det}(p_j - p_i, p_k - p_i) \in \{-1, +1\}.$$

The map χ is alternating. We store only x_{ijk} for $i < j < k$, with $x_{ijk} = 1$ meaning $\chi(i, j, k) = +1$.

After a generic rotation, no two points have the same first coordinate. Relabeling therefore allows us to assume

$$(2.1) \quad (p_1)_x < (p_2)_x < \dots < (p_n)_x.$$

This normalization loses no geometric configurations.

We use the following elementary four-set criterion.

Lemma 2.1 (Four-set criterion). *A finite general-position planar set S is in convex position if and only if every four-element subset of S is in convex position.*

Proof. Every subset of a set in convex position is in convex position. Conversely, if S is not in convex position, some $p \in S$ lies in the convex hull of $S \setminus \{p\}$. Triangulate the convex hull using vertices of $S \setminus \{p\}$. The point p belongs to one of the closed triangles of the triangulation. General position excludes membership in an edge, so p is in the interior of a triangle with vertices $a, b, c \in S \setminus \{p\}$. Thus $\{a, b, c, p\}$ is not in convex position. $\square$

3. THE FOUR-POINT PARITY THEOREM

The next theorem is independent of the ordering in (2.1). Its left-hand side is invariant under a relabeling of the four points because every transposition changes the signs of exactly two of the four factors.

Theorem 3.1 (Orientation parity). *Let a, b, c, d be four points in general position. They are in convex position if and only if*

$$(3.1) \quad \chi(a, b, c)\chi(a, b, d)\chi(a, c, d)\chi(b, c, d) = +1.$$

Equivalently, after identifying a positive sign with Boolean value one, a nonconvex four-set has odd orientation parity.

Proof. Append a final coordinate 1 to each point and form the 3×4 matrix with columns $(a, 1), (b, 1), (c, 1), (d, 1)$. Cofactor expansion gives a nonzero vector in its nullspace whose coefficient signs, in the displayed order, are

$$(3.2) \quad (-\chi(b, c, d), \chi(a, c, d), -\chi(a, b, d), \chi(a, b, c)).$$

No coefficient vanishes by general position. The positive and negative supports in this unique affine dependence form the Radon partition of the four points. The points are in convex position exactly when that partition

has type $2 + 2$; the alternative type $1 + 3$ says that one point lies in the triangle of the other three.

Write the four signs in (3.1) as s_1, s_2, s_3, s_4 . The product of the four coefficient signs in (3.2) is $s_1 s_2 s_3 s_4$. Four nonzero real numbers have two positive and two negative entries exactly when their product is positive, once the impossible all-positive and all-negative affine dependences are excluded. Those two cases are impossible because the affine coefficients sum to zero. Thus the Radon partition has type $2 + 2$ exactly when (3.1) is positive. $\square$

This proof also indicates why a one-bit summary is sufficient: for the purpose of excluding convex k -sets, the identity of a possible interior point of a four-set is immaterial.

4. THE RANK-THREE SIGNOTOPE RULE

Signotopes give a convenient local abstraction of planar order types [6]. We record the required implication directly so that the soundness argument is self-contained.

Lemma 4.1 (One-change rule). *Under the order (2.1), for every $a < b < c < d$ the sign sequence*

$$(4.1) \quad \chi(b, c, d), \chi(a, c, d), \chi(a, b, d), \chi(a, b, c)$$

has at most one sign change.

Proof. Put $A = x_b - x_a$, $B = x_c - x_b$, and $C = x_d - x_c$, all positive, and let α, β, γ be the slopes of ab, bc, cd , respectively. Set $u = \beta - \alpha$ and $v = \gamma - \beta$. Positive rescaling shows that the four signs in (4.1) are the signs of

$$(4.2) \quad v, \quad v + \frac{A}{A+B}u, \quad u + \frac{C}{B+C}v, \quad u.$$

If u and v have the same sign, all four expressions have that sign. If they have opposite signs, two separated reversals in (4.2) would require, after possibly multiplying all expressions by -1 ,

$$\frac{v}{-u} < \frac{A}{A+B} < 1 \quad \text{and} \quad \frac{v}{-u} > \frac{B+C}{C} > 1,$$

which is impossible. Hence at most one change occurs. $\square$

For $a < b < c < d$, abbreviate

$$(u, v, w, t) = (x_{abc}, x_{abd}, x_{acd}, x_{bcd}).$$

The standard ordered signotope CNF [12] is

$$(4.3) \quad (\neg u \vee \neg w \vee v) \wedge (u \vee w \vee \neg v),$$

$$(4.4) \quad (\neg u \vee \neg t \vee w) \wedge (u \vee t \vee \neg w).$$

The first pair says that if $u = w$ then v has their common value; the second says that if $u = t$ then w has their common value. A truth-table check, or a short case split on the first change, shows that (4.3)–(4.4) accept exactly the

eight sequences with at most one change. Thus they are equivalent to the direct encoding by eight four-literal forbidden-pattern clauses.

Proposition 4.2 (Clause minimality). *Without auxiliary variables, four clauses are necessary and sufficient to define the one-change Boolean relation on four signs.*

Proof. There are eight accepted and eight rejected assignments. No non-tautological implicate involving at most two variables is valid on all accepted assignments: any prescribed values at one or two positions extend to a one-change sequence (use a constant extension when the values agree and put the sole change between them when they differ). Hence every clause in an equivalent CNF contains at least three distinct variables and can exclude at most two of the rejected assignments. At least four clauses are therefore required, and (4.3)–(4.4) attain the bound. The release verifier independently enumerates all $3^4 - 1$ nonempty, non-tautological clauses and checks the same minimum as an exact set-cover calculation. $\square$

These axioms are necessary for realizable point sets but not sufficient for realizability. The eight-clause direct form remains in the generator solely as a controlled propagation benchmark.

5. A PARITY-COMPRESSED REFUTATION ENCODING

Fix integers $n \geq k \geq 4$. The formula $F_{n,k}$ has an orientation variable x_{ijk} for every triple $i < j < k$, and a variable z_Q for every four-set Q . Its literal Boolean meaning is

$$(5.1) \quad z_Q = 1 \implies Q \text{ has odd orientation parity.}$$

For realizable orientation assignments, Theorem 3.1 identifies this with nonconvexity. No such geometric interpretation is assigned to an arbitrary nonrealizable SAT model. It consists of the following blocks.

Signotope block. For each $Q = \{a < b < c < d\}$, add the four ternary clauses (4.3)–(4.4).

Parity-guard block. Under the one-change rule, odd parity has a useful factorization:

$$(5.2) \quad u \oplus v \oplus w \oplus t = 1 \iff (u \neq t) \wedge (v = w).$$

The same endpoint/middle split is implicit in the two convex-quadrilateral cases of Koshelev and Koshka [8]; (5.2) is their complement in the uncolored, no-density setting. We make no priority claim for the factorization itself. Indeed, equal endpoints mean zero changes and hence even parity. With unequal endpoints there is exactly one change; a middle change gives two zeros and two ones, whereas an outer change gives odd parity, exactly when $v = w$. Consequently (5.1) is encoded by

$$(5.3) \quad \begin{aligned} & (\neg z_Q \vee u \vee t) \wedge (\neg z_Q \vee \neg u \vee \neg t), \\ & (\neg z_Q \vee \neg v \vee w) \wedge (\neg z_Q \vee v \vee \neg w). \end{aligned}$$

These are four ternary clauses. The reverse implication is deliberately not imposed. For comparison, enumerating the four admissible even patterns gives four five-literal clauses; the release verifies that the two forms are equivalent jointly with the signotope block.

For a strongest local comparison, full equivalence can also be factored into eight ternary clauses. In addition to (5.3), its reverse half is

$$(z_Q \vee \neg w \vee t) \wedge (z_Q \vee w \vee \neg t), \\ (z_Q \vee \neg u \vee v) \wedge (z_Q \vee u \vee \neg v).$$

Proposition 5.1 (Local reification minima). *Modulo the one-change relation and without auxiliary variables, a CNF for $z_Q \Rightarrow \text{odd parity}$ needs at least four clauses, while a CNF for $z_Q \Leftrightarrow \text{odd parity}$ needs at least eight. The factored forms above attain both bounds using only ternary clauses.*

Proof. This is a finite exact-cover calculation included in the release. Enumerate the $3^5 - 1 = 242$ nonempty clauses in which each of z_Q, u, v, w, t is absent, positive, or negative. Discard clauses that reject an accepted assignment, and record which rejected one-change assignments are excluded by each remaining clause. Exhaustive set cover has optimum four for the implication relation and eight for equivalence. Direct truth-table evaluation verifies that the displayed clauses attain those optima. The checker performs this enumeration afresh and does not trust a stored solver answer. $\square$

Seven-set exclusion block. For every $K \in \binom{[n]}{k}$, add the clause

$$(5.4) \quad \bigvee_{Q \in \binom{K}{4}} z_Q.$$

This clause has width $\binom{k}{4}$.

Theorem 5.2 (Refutation-sound relaxation). *Every general-position n -point set with no k points in convex position induces a satisfying assignment of $F_{n,k}$. Consequently,*

$$F_{n,k} \text{ is UNSAT} \implies \text{ES}(k) \leq n.$$

The converse implication from SAT to a realizable counterexample is not asserted.

Proof. Take a geometric counterexample, rotate it generically, and label it as in (2.1). Assign every x_{ijk} its orientation value. The signotope block is satisfied by Lemma 4.1. For each four-set Q , set $z_Q = 1$ exactly when Q is nonconvex. The parity guards are satisfied by Theorem 3.1.

It remains to check (5.4). A k -set K in the assumed point set is not in convex position. Lemma 2.1 supplies a nonconvex four-subset $Q \subset K$, for which $z_Q = 1$. Thus the clause for K is satisfied. Every geometric counterexample therefore maps to a Boolean model, so the contrapositive proves the stated UNSAT implication. $\square$

Remark 5.3 (Why one-sided reification is enough). The formula never needs to infer $z_Q = 1$ from odd parity. For refutation-soundness we only need the canonical extension used in the proof. Within a realizable assignment, the one-sided form prevents a convex four-set from serving as a witness in (5.4), while avoiding the clauses and variables needed for full reification.

6. EXACT SIZE AT $(n, k) = (33, 7)$

There are $\binom{33}{3} = 5,456$ orientation variables and $\binom{33}{4} = 40,920$ indicators. The three clause blocks have sizes

$$(6.1) \quad 4 \binom{33}{4} = 163,680, \quad 4 \binom{33}{4} = 163,680, \quad \binom{33}{7} = 4,272,048.$$

The first two blocks have width three; every clause in the last block has width $\binom{7}{4} = 35$.

Proposition 6.1 (Audited formula counts). *The formula $F_{33,7}$ has exactly*

$$\boxed{46,376 \text{ variables, } 4,599,408 \text{ clauses, } 150,503,760 \text{ literal occurrences.}}$$

Proof. The variable and clause totals follow from (6.1). Counting literal occurrences by clause width gives

$$3 \cdot 163,680 + 3 \cdot 163,680 + 35 \cdot 4,272,048 = 150,503,760.$$

The accompanying verifier recomputes these expressions from the parameters, rather than trusting serialized totals. $\square$

Table 1 compares five encodings. The first row is our audit of the public 2025 generator before convex-layer anchors are selected: it uses 5,456 triple variables and 14 selector variables per four-set. The second row keeps that generator’s reduced five-point implication block and replaces only its four-set selectors by the unrestricted eight-clause parity guard. This is a drop-in refutation-sound relaxation for its anchored searches. The third row audits the fixed-order specialization of the public implementation of Subercaseaux et al.: one convexity variable, sixteen five-literal direct full-reification clauses, and four signotope clauses per four-set. The actual emitted CNF, rather than the prose description alone, is the basis of this row; the public $n = 9$ formula retaining generic order variables is benchmarked separately below. The fourth row retains our direct eight-clause signotope block as a propagation control, while the final row is the formulation proved above.

Relative to the audited public-generator baseline, the last row reduces the variable count by 91.9811%, the clause count by 72.4104%, and literal occurrences by 88.0662%. Relative to the fixed-order public full reification, the reductions are 0.0000%, 9.6463%, and 1.8153%, respectively. Formula size is not a prediction of solver time: propagation, learned-clause quality, symmetry breaking, and the branching distribution may dominate. Moreover, convex-layer labels are cyclic hull labels, whereas the smallest encoding fixes

TABLE 1. Exact unanchored core counts at (33, 7). “Literals” means literal occurrences, not distinct literals.

Encoding	Variables	Clauses	Literals
2025 public-generator baseline	578,336	16,670,808	1,261,154,400
Anchor-compatible half-reification	46,376	14,092,848	208,119,120
Public full reification, fixed order	46,376	5,090,448	153,286,320
Factored full reification, fixed order	46,376	4,763,088	150,994,800
Direct signotope, factored guard	46,376	4,763,088	151,322,160
Four-clause signotope, factored guard	46,376	4,599,408	150,503,760

global x -order. Porting all existing anchors is therefore a modeling task, not a syntactic substitution.

The closest optimized local control is the factored full-reification row, not the public direct truth-table row. Against that control, one-sided reification saves 163,680 clauses (3.4364%) and 491,040 literal occurrences (0.3252%), with no variable reduction. This is the incremental size gain attributable specifically to dropping the reverse implication.

6.1. Reproduction boundary for the public full-reification row. The prose description of the Subercaseaux–Mackey–Qian–Heule encoding gives a $12\binom{n}{4}$ Tseitin compilation. Introducing two intermediate variables for the three nested equivalences indeed gives twelve ternary clauses per four-set. The public generator snapshot audited here emits a different CNF: it creates only the variable `conv`, passes the complete five-variable equivalence to EZNF, and thereby obtains sixteen five-literal truth-table clauses with no parity auxiliaries. Thus the third row of Table 1 compares with the implementation actually emitted by the public code, not with the alternative Tseitin compilation described in the text.

The source is the public `automatic-symmetries` repository at <https://github.com/bsubercaseaux/automatic-symmetries>; the exact revision, environment, and source locations are recorded in the companion reproduction note. From the repository root the audited command is

```
python3 encoders/erdos_szekeres.py -n 9 -g 5 -s 0.
```

Because the upstream repository has no dependency lock, our record pins Python 3.9.6, EZNF 0.120.2, and Lark 1.3.1. A clean temporary-directory run reproduced the retained CNF byte for byte. An independent parser checks its 282 variables, 7,254 clauses, width histogram, all $16\binom{9}{4} = 2,016$ direct full-reification clauses, and the absence of parity auxiliary variables.

7. VERIFICATION AND ADVERSARIAL TESTS

The release contains a streaming DIMACS generator and six logically separated checks.

- (1) A direct integer-grid enumerator compares convex-hull size with the orientation product for every general-position four-subset it visits.
- (2) An independent cofactor program checks the sign pattern of (3.2) and the Radon partition without calling the convex-hull routine.
- (3) A Boolean truth-table test proves equivalence of the four- and eight-clause signotope encodings, verifies the factored guard, and computes the clause-minimality cover. A separate counter recomputes all entries of Proposition 6.1.
- (4) A six-point enumerator checks all 908 signotopes and verifies that the eight direct relations recognize exactly the 16 assignments whose fifteen four-subsets all have even parity.
- (5) A certificate verifier re-parses the released (9, 5) DIMACS file, regenerates its clauses exactly from the current generator, reruns DRAT-trim, and separately reruns an LRAT checker.
- (6) A 17-point verifier reconstructs the 61-cube cover, confirms the exact base-plus-units form of every leaf, checks all 61 DRAT proofs, and separately checks all 61 converted LRAT proofs.

The current release checks 1,278 general-position grid quadruples by the first route and 9,498 by the second. These finite tests are not substitutes for the proofs above; they are regression tests designed to expose sign conventions, clause polarity errors, and count drift.

7.1. A checked end-to-end certificate. For $(n, k) = (9, 5)$, the recommended formula has 210 variables, 1,134 clauses, and 3,654 literal occurrences. Kissat 4.0.4 returned UNSAT after 2,646 conflicts and 7,525 decisions, producing a 36,761-byte binary DRAT trace. DRAT-trim accepted the trace and converted its proof core to a 183,964-byte LRAT file; the independent LRAT checker also returned VERIFIED. Together with the classical lower bound, this reproduces the known equality $ES(5) = 9$. It is a certificate for the release pipeline, not a new mathematical bound.

Table 2 records same-machine Kissat runs on an Apple M3 Pro laptop with 36 GB memory. In addition to the 2025 public baseline, we ran the public generator of Subercaseaux et al.; its generic order-variable formula has 282 variables and 7,254 clauses. Neither public formula terminated within the ten-second bound, while the three fixed-order signotope variants finished in under a tenth of a second. The variable counts in table order are 1,848, 282, 210, 210, and 210. The corresponding learned-clause counts are 127,907, 327,086, 2,642, 2,682, and 2,537. Proof logging was disabled in the timing runs. Separate proof-producing runs for the three terminating variants gave binary DRAT traces of 57,084, 38,376, and 36,761 bytes, respectively, each accepted by DRAT-trim; the last was additionally converted to and checked as LRAT. No complete trace exists for either timed-out row.

Since this is one small instance and runtime noise is substantial at that scale, we draw no asymptotic speedup claim. In particular, the factored full indicator is slightly faster here. The supported conclusion is only that the

TABLE 2. Kissat 4.0.4 on the known (9, 5) UNSAT instance.
Time is wall time; memory is peak resident memory measured
by the operating system.

Encoding	Clauses	Status	Time (s)	MiB	Conflicts	Decisions
Public-generator baseline	14,112	UNKNOWN	10.02	19.4	129,593	378,943
SMQH public generator	7,254	UNKNOWN	10.01	23.0	329,901	794,808
Direct signotope, half	1,638	UNSAT	0.05	3.0	2,802	8,843
Factored full indicator	1,638	UNSAT	0.04	3.0	2,817	6,020
This work, factored half	1,134	UNSAT	0.05	2.9	2,646	7,525

one-sided size reduction causes no evident propagation failure on this test; an actual speed advantage has not been established.

As an adversarial benchmark, we also tested the reduced five-point implication family used by the public 2025 generator without its anchors. Combined with the parity block it admits a Boolean model at the known (17, 6) threshold. This does not contradict $ES(6) = 17$: it shows that the reduced unanchored formula is a relaxation and that its SAT models can be pseudomodels. The x -ordered signotope version gives the expected SAT/UNSAT answers at (8, 5) and (9, 5) and a SAT answer at (16, 6).

7.2. The exact status of the 17-point signotope test. The status of the (17, 6) formula is not an open realizability question. Szekeres and Peters define the same eight allowed four-sign patterns as our ordered one-change relation and prove that every such signature function on 17 labels contains a convex six-subset [13, Theorem 2]. Their theorem ranges over nonrealizable signature functions as well as point configurations. It therefore settles the abstract formula itself.

Proposition 7.1. *The unanchored one-sided formula $F_{17,6}$ is unsatisfiable.*

Proof. Suppose that $F_{17,6}$ has a model. Its orientation variables define a signature function in the class Σ_{17}^* of Szekeres and Peters, because (4.3)–(4.4) accept exactly their eight patterns. Their theorem provides a six-set K whose every four-subset is convex. By their four-point classification, equivalently by Theorem 3.1, each $Q \in \binom{K}{4}$ has even orientation parity. The guards (5.3) consequently force every z_Q with $Q \subset K$ to zero. This falsifies (5.4), a contradiction. $\square$

For an independent local check of the identification, the release also implements the eight four-term convex-six relations in equation (4.1) of Szekeres and Peters. Exhaustive enumeration finds 908 labeled signotopes on six points; exactly 16 have all fifteen four-subsets of even parity, and the direct relations recognize exactly those same 16 assignments. Applying the relations to every six-set gives a 17-point CNF with 680 variables, 207,537 clauses, and 820,625 literal occurrences after one global orientation unit.

We then carried the geometric normalization through a complete certificate. Choose a hull vertex as label 1, order the other points counterclockwise

around it, and apply the order-type-preserving transformation of Heule and Scheucher so that the labels also increase in x -coordinate [7, Section 3.4]. This makes all $\binom{16}{2} = 120$ orientations x_{1ab} positive. Adding the four missing ternary prime implicates of the one-change relation for each four-set does not change its models. Explicitly, for $(p, q, r, s) = (x_{abc}, x_{abd}, x_{acd}, x_{bcd})$, they are

$$(\neg q \vee r \vee \neg s), \quad (\neg p \vee q \vee \neg s), \quad (p \vee \neg q \vee s), \quad (q \vee \neg r \vee s).$$

The release truth table verifies that these are consequences of (4.3)–(4.4). The resulting direct-relation base has 680 variables, 217,176 clauses, and 849,304 literal occurrences.

For the first six labels, ten orientation variables remain after the extreme units. Of their 1,024 assignments, 963 already falsify a base clause and 61 remain. Appending each remaining assignment as a cube gives an exhaustive, pairwise-disjoint partition: every base model restricts to exactly one of the 61 cubes. Kissat 4.0.4 returned UNSAT on every leaf. The sum of the 61 single-thread wall times was 1,767.817 seconds; the median was 23.912 seconds and the maximum was 100.976 seconds. The 61 binary DRAT files contain 288,562,058 bytes in total. A separate verifier reconstructs the 1,024-case coverage argument, checks that every leaf is the exact base plus its ten unit clauses, and reruns DRAT-trim on all 61 proofs; every proof was accepted. DRAT-trim also converted the leaves to 871,234,978 bytes of LRAT, and the separate LRAT checker accepted all 61 converted files. Thus the release contains a complete checked certificate for the normalized geometric 17-point test, rather than only a bounded solver experiment.

The normalized certificate is not a refutation of the unanchored abstract formula, whereas Proposition 7.1 is a mathematical consequence of the broader Szekeres–Peters theorem. Bounded monolithic runs of our one-sided and direct-relation CNFs ended UNKNOWN: six 180-second one-sided runs, four 300-second prime-implicate runs, and three 900-second runs produced no terminating trace. These failures do not weaken the proposition; they show that fewer clauses alone do not reproduce the propagation strength of specialized encodings. Indeed, Heule and Scheucher report 17.39 seconds for their optimized symmetry-broken formula and 8.53 seconds after adding a partition into 1,108 cubes in iCNF format [7]. We do not claim that the extreme-point units preserve every abstract signotope isomorphism class.

No solver trace for $F_{33,7}$ is included. A future claim that the formula is UNSAT should retain the exact DIMACS instance, emit a standard proof trace, and validate that trace with an independently implemented checker; standard proof-logging practice is described, for example, by Wetzler, Heule, and Hunt [14].

8. WHAT REMAINS TO DETERMINE ES(7)

The lower bound in (1.1) is already sharp if and only if the unanchored 33-point counterexample space is empty. The present work moves that question

to a smaller, exactly specified refutation-sound Boolean relaxation, but it leaves two mutually exclusive completion routes.

- *Proof route.* Refute $F_{33,7}$, or a sound strengthening of it, and independently check a complete proof trace. Since every geometric counterexample satisfies $F_{33,7}$, such a refutation proves $\text{ES}(7) \leq 33$ and hence $\text{ES}(7) = 33$.
- *Disproof route.* Produce 33 rational coordinate pairs, verify every triple determinant is nonzero, and verify exactly that no seven-set is in convex position. A bare SAT assignment is insufficient because signotope axioms do not guarantee point realizability.

Useful next computational steps include certified decomposition into symmetry breaking cubes, clause-learning experiments that preserve proof logging, and the addition of sound order-type constraints only when their complete CNF derivations are audited. Solver performance data should be kept separate from mathematical certificates.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The companion directory contains the deterministic formula generator and independent verification programs. It also contains the count certificate, the checked (9, 5) DRAT/LRAT traces, the 61-leaf (17, 6) DRAT/LRAT family and its cube manifest, solver logs, audit ledgers, and the manuscript source. Detailed 17-point regeneration commands are in the companion reproduction note. From the project root, the verification entry point is `python3 code/run_all_verifiers.py`.

AI-assisted tools were used for exploratory derivations, code drafting, and literature organization. The mathematical claims retained in the paper were then reduced to the displayed proofs and deterministic verification scripts. The authors remain responsible for the arguments, source checking, and any future submission. The computational experiments reported here did not produce the missing global certificate.

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